



BIPOLAR DISORDER



Bipolar Disorder

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- Bipolar disorder is a **mood disorder** characterized by one or more episodes of **mania** or **hypomania**, often with **a history of one or more MDD episodes**.
- It is a chronic and progressive illness, previously known as ***manic-depressive illness***, *is a cyclical*, lifelong disorder with recurrent extreme fluctuations in **emotion, mood, energy, and behavior**.
 - Mood episodes can be manic, depressed, or mixed with or without psychosis

Symptom Domains of Bipolar Disorder

Manic mood and behavior

- Euphoria
- Grandiosity
- Pressured speech
- Impulsivity
- Excessive libido
- Recklessness
- Social intrusiveness
- Diminished need for sleep

Dysphoric or negative mood and behavior

- Depression
- Anxiety
- Irritability
- Hostility
- Violence
- Suicide

BIPOLAR DISORDER

Psychotic symptoms

- Delusions
- Hallucinations

Cognitive symptoms

- Racing thoughts
- Distractibility
- Disorganization
- Inattentiveness

Etiology and Pathophysiology

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- **The precise etiology is unknown.**
- Bipolar disorder is influenced by a variety of factors that may enhance gene expression

Medical conditions that induce mania

- CNS disorders (brain tumor, strokes, head injuries)
- Infections (encephalitis, neurosyphilis, sepsis, HIV)
- Electrolyte or metabolic abnormalities (calcium or sodium fluctuations, hyper- or hypoglycemia)
- Endocrine or hormonal dysregulation

Medications/drugs that induce mania

- ✚ Alcohol intoxication
- ✚ **Antidepressants** (MAOIs, TCAs, 5-HT and/or NE and/or DA reuptake inhibitors, 5-HT antagonists)
- ✚ Marijuana intoxication precipitates psychosis, paranoid thoughts, anxiety, and restlessness
- ✚ **Steroids** (anabolic, adrenocorticotrophic hormone, corticosteroids)

Clinical presentation and diagnosis

Mood Disorders Defined by Episodes

Disorder Subtype	Episode(s)
MDD, single episode	Major depressive episode
MDD, recurrent	Two or more major depressive episodes
Bipolar I disorder	Manic episode ± MDD or hypomanic episode
Bipolar II disorder	MDD episode + hypomanic episode
Persistent depressive disorder (Dysthymia)	Depressed mood most days for at least 2 yrs.
Cyclothymic disorder	Chronic fluctuations between subsyndromal depressive and hypomanic episodes (2 years for adults)
Rapid cycling	At least four episodes of mania, hypomania, or depression in 1 year
Unspecified bipolar and related disorder	Mood states do not meet full criteria for any specific disorder in bipolar and related disorders class

Major Depressive Episode

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- In bipolar depression, pts often *have mood lability, hypersomnia, low energy, psychomotor retardation, cognitive impairments (trouble remembering, learning new things, concentrating, or making decisions), anhedonia, decreased sexual activity, slowed speech, carbohydrate craving, and weight gain.*
- ***Delusions, hallucinations, and suicide attempts*** are more common in bipolar depression than in unipolar depression.

Manic Episode

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- Acute mania usually **begins abruptly**, and symptoms increase over several days.
 - ▣ The severe stages may include bizarre behavior, hallucinations, and paranoid or grandiose delusions.
- There is marked impairment in functioning or need hospitalization.
 - ▣ Manic episodes may be precipitated by stressors, sleep deprivation, antidepressants, CNS stimulants, or bright light.

DSM-V Criteria for a Manic Episode

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1. A distinct period of abnormally and persistently elevated, expansive, or irritable mood, lasting ≥ 1 week (or of any duration if hospitalization is necessary).
2. During the period of mood disturbance, ≥ 3 of the following symptoms have persisted (4 if the mood is only irritable) and have been present to a significant degree:
 - Inflated self-esteem or grandiosity
 - Decreased need for sleep (e.g., feels rested after only 3 hours of sleep)
 - More talkative than usual or pressure to keep talking
 - Flight of ideas or subjective experience that thoughts are racing
 - Distractibility (i.e., attention too easily drawn to unimportant or irrelevant external stimuli)
 - Increase in goal-directed activity (either social, at work, at school, or sexually) or psychomotor agitation
 - Excessive involvement in pleasurable activities that have a high potential for painful consequences (e.g., the person engages in unrestrained buying sprees, sexual indiscretions, or foolish business investing)
3. The symptoms do not meet the criteria for a *Mixed Episode*.
4. The mood disturbance is sufficiently severe to cause marked impairment in occupational functioning or in usual social activities or relationships with others, or to necessitate hospitalization to prevent harm to self or others, or there are psychotic features.
5. The symptoms are not due to the direct physiologic effects of a substance (e.g., a drug of abuse, a medication, or other treatment) or a general medical condition (e.g., hyperthyroidism).

DIGFAST (manic episode)

- **D**istractable (poorly focused)
- **I**ndiscretion (excessive pleasurable activities)
- **G**randiosity (unrealistic belief in one's abilities)
- **F**light of Ideas
- **A**ctivities (increased goal direction)
- **S**leep deficit (decreased need for sleep)
- **T**alkativeness (pressured speech)

Hypomanic Episode

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- There is **no marked impairment** in social or occupational functioning, no delusions, and no hallucinations.
- During a hypomanic episode, some pts may be more **productive and creative than usual**, but 5% to 15% of pts may rapidly switch to a manic episode.

Mixed Episode

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- The presence of high and low symptoms occurring at the same time
 - ▣ *Mixed episodes occur in up to 40% of all episodes, are often difficult to diagnose and treat, and are more common in younger and older pts and females.*
- Pts with mixed states often have comorbid alcohol and substance abuse, severe anxiety symptoms, a higher suicide rate, and a poorer prognosis.

Course of Illness

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- The average age of onset of a first manic episode is 21 yrs.
 - ▣ More than 80% of bipolar pts have more than four episodes during their lifetime.
- Usually there is normal functioning between episodes.
- **Rapid cyclers** (10% to 20% of bipolar pts) have **four or more episodes per yr** (major depressive, manic, mixed, or hypomanic).

- **Rapid-cycling** and **mixed states** are associated with a poorer prognosis and non-response to anti-manic agents.
- **Risk factors for rapid cycling include:**
 - Biologic rhythm dysregulation,
 - Antidepressant or stimulant use,
 - Hypothyroidism, and
 - Premenstrual and postpartum states.

- Women are more likely to have mixed states, depressive episodes, and rapid cycling than men.
- **Suicide attempts occur in up to 50% of pts with bipolar disorder**
 - **Bipolar II pts may be more likely than bipolar I pts to attempt suicide.**

Treatment

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Goals of treatment

- Eliminate mood episode with complete remission of symptoms (i.e., **acute treatment**)
- Prevent recurrences or relapses of mood episodes (i.e., **continuation phase treatment**)
- Return to complete psychosocial functioning
- Maximize adherence with therapy
- Minimize adverse effects

- Treat co-morbid substance use and abuse
- *Eliminate alcohol, marijuana, cocaine, amphetamines, and hallucinogens*
- *Minimize nicotine use and stop caffeine intake at least 8 hours prior to bedtime*
- Avoidance of stressors or substances that precipitate an acute episode

Non-Pharmacologic Treatment

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- ▣ **Psychotherapy** (individual, group, and family), interpersonal therapy, and/or cognitive behavioral therapy
- ▣ **Stress Reduction** techniques, relaxation therapy, massage, yoga
- ▣ **Sleep** (regular bedtime and awake schedule; avoid alcohol or caffeine intake prior to bedtime)
- ▣ **Nutrition** (regular intake of protein-rich foods or drinks and essential fatty acids; supplemental vitamins and minerals)
- ▣ **Exercise**

Pharmacologic Therapy

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- For **acute mania** in bipolar disorder
 - ▣ Lithium, divalproex sodium (valproate), aripiprazole, olanzapine, quetiapine, risperidone, and ziprasidone
 - ▣ BDZ
- **Maintenance** treatment of bipolar disorder.
 - ▣ Lithium, divalproex sodium (sodium valproate), olanzapine, and lamotrigine

- **Quetiapine** is the only antipsychotic that is FDA approved for **bipolar depression**
- **Lithium is the drug of choice for bipolar disorder with euphoric mania**
- **Valproate** has better efficacy for mixed states, irritable/dysphoric mania, and rapid cycling compared with lithium

Combination therapies

- Lithium plus valproate or carbamazepine;
- Lithium or valproate plus an atypical antipsychotic may provide better acute response and prevention of relapse and recurrence than monotherapy in some bipolar pts, especially those with **mixed states or rapid cycling**.

Lithium

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- **Lithium** was the first established mood stabilizer and is still considered a first-line agent for **acute mania and maintenance treatment of both bipolar I and II disorders.**
- **Long-term use of lithium reduces suicide risk.**

Cont.....

- MOA: is unknown, but it appears to be neuroprotective.
 - ▣ Some ideas include attenuation of dopamine function; facilitation of GABA function
 - ▣ Alteration of serotonin function, such as improvement in 5HT-1A and reduction in 5HT-2 neurotransmission.
- **Lithium is effective for manic and depressive components.**
 - ▣ Although it is **not a particularly good antidepressant** as monotherapy in **unipolar depression**

Cont.....

- Anti-manic effects can occur in **1–2 weeks**.
 - ▣ Most clinicians **use antipsychotics and/or benzodiazepines** as adjunctive therapy during this period to cover the agitation and other symptoms

Pharmacokinetics:

- ▣ Its half-life is 20–24 hours.
- ▣ It is excreted 95% unchanged by kidney, and anything that alters GFR affects its clearance.

Cont.....

- **Initial dosing is in the range of**
 - ▣ 600–900 mg/day in divided doses and then titrated according to response and tolerability.
 - ▣ Maintenance doses are based on serum concentrations, symptom relief, and the occurrence of adverse effects.
- **A pre-lithium workup includes**
 - ▣ A complete blood cell count, electrolytes, renal function, thyroid function tests, urinalysis, ECG, and pregnancy test for women of childbearing age.

Lithium side effect: **LITH**

- **L**eukocytosis
- **I**nsipidus (Nephrogenic Diabetes Insipidus)
- **T**remor/**T**eratogenic (Ebstein's anomaly)
- **H**ypothyroidism



Adverse Effects Associated with Lithium

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Problem	Potential Interventions
Rash or ↑ psoriasis	Discontinue the drug temporarily or permanently
Tremor	Reduce dose; add β -blocker
CNS toxicity (e.g., agitation, confusion)	Reduce dose
GI (nausea, vomiting, diarrhea)	Reduce dose; try ER product, split doses
Hypothyroidism	Give levothyroxine; discontinue lithium, if necessary
Polydipsia or polyuria	Reduce dose, manage fluid intake; single bedtime dosing helps
Interstitial fibrosis, glomerulosclerosis	Keep dose at lowest effective concentration; avoid dehydration
Teratogenicity	Avoid during first trimester, if possible

Cont.....

□ **Monitoring:**

- Serum concentrations must be monitored.
- Half-life is about 1 day, so steady state occurs in about 5 days.
- Most clinicians will aim for concentrations of **0.8–1.2** mEq/L in acute mania and **0.6–1.0** mEq/L during maintenance.
- Concentration-response data are based on 12-hour post dose concentrations
- Perform **renal function tests, thyroid function tests, and a urinalysis** every 6–12 months.

- Combining lithium with typical AP may cause **neurotoxicity**
- **GI distress** may be minimized by standard approaches or by adding antacids or antidiarrheals.
- **Muscle weakness** and transient lethargy occur in about 30% of pts.

Anticonvulsant

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- **Valproate is as effective as lithium and olanzapine for pure mania**
 - ▣ It can be more effective than lithium for rapid cycling, mixed states bipolar disorder.
 - ▣ It reduces the frequency of recurrent manic, depressive, and mixed episodes.
- Valproate can be added to lithium or carbamazepine to achieve synergistic effects.

Anticonvulsant

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- Carbamazepine is usually reserved for lithium-refractory pts, rapid cyclers, or mixed states.
- The combination of carbamazepine with **lithium, valproate, and AP** is often used for manic episodes in treatment-resistant pts.

Anticonvulsant

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- **Lamotrigine is effective for maintenance treatment of bipolar I disorder in adults.**
 - ▣ It has been used as monotherapy or add-on therapy for refractory bipolar depression.
- **Beneficial for** maintenance therapy of treatment-resistant bipolar I and II disorders, rapid cycling, and mixed states.
- Better safety profile

Antipsychotics

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- **Atypical antipsychotic drugs** (Aripiprazole, olanzapine, quetiapine, risperidone, and ziprasidone) are effective as **monotherapy** or as **add-on therapy** to lithium or valproate for acute mania.
- **Prophylactic use** of antipsychotics can be needed for some pts with recurrent mania or mixed states
- **Both typical and atypical antipsychotics are effective in approximately 70% of pts with acute mania associated with agitation, aggression, psychosis**

- Only **olanzapine** is FDA approved for **maintenance** treatment
- SGAs have demonstrated similar efficacy for treatment of acute mania
 - ▣ associated with agitation, aggression, and psychosis
- **Clozapine** monotherapy has acute and long-term mood stabilizing effects in **refractory bipolar disorder**, including mixed mania and rapid cycling

Benzodiazepines

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- High-potency **benzodiazepines** (e.g., **clonazepam** and **lorazepam**) are common alternatives to or in combination with antipsychotics for
 - ▣ Acute mania, agitation, anxiety, panic, and insomnia or in those who cannot take mood stabilizers
 - ▣ **Lorazepam IM** may be used for acute agitation
 - Benefits of lorazepam include availability of both IM and oral formulations (tablet and concentrate), lack of active metabolites, and safety in hepatic and renal impairment

Antidepressants

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- **Antidepressants (SSRIs)** are routinely added for the treatment of **acute depression**
- **TCAs** are associated with an **increased risk of inducing mania in bipolar I disorder** and possibly cause rapid cycling.

Special Populations

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- Approximately 20% to 50% of women with bipolar disorder relapse postpartum; prophylaxis with mood stabilizers (e.g., **lithium or valproate**) is recommended immediately postpartum to decrease the risk of relapse.
- When **valproate** is taken during the first trimester, the risk of neural tube defect is 5% to 9%.
 - ▣ It is contraindicated during pregnancy
 - ▣ Administration of folic acid can reduce this risk

Summary

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Acute Manic or Mixed Episode

General Guidelines

- Assess for secondary causes of mania or mixed states (eg alcohol or drug use)
- Discontinue antidepressants
- Taper off stimulants and caffeine if possible
- Treat substance abuse
- Encourage good nutrition (with regular protein and essential fatty acid intake), exercise, adequate sleep, stress reduction, and psychosocial therapy

Hypomania

- First**, optimize current mood stabilizer or initiate mood-stabilizing medication: lithium,^a valproate,^a carbamazepine,^a or SGAs
- Consider adding a benzodiazepine (lorazepam or clonazepam) for short-term adjunctive treatment of agitation or insomnia if needed
- Alternative medication treatment options: oxcarbazepine

Mania

- First**, two- or three-drug combinations (lithium,^a valproate,^a or SGA) **plus** a benzodiazepine (lorazepam or clonazepam) and/or antipsychotic for short-term adjunctive treatment of agitation or insomnia; lorazepam is recommended for catatonia
- Do not combine antipsychotics
- Alternative medication treatment options: carbamazepine^a; if patient does not respond or tolerate, consider oxcarbazepine

Second, if response is inadequate, consider a two-drug combination:

- Lithium^a **plus** an anticonvulsant or an SGA
- Anticonvulsant **plus** an anticonvulsant or SGA

Second, if response is inadequate, consider a three-drug combination:

- Lithium^a **plus** an anticonvulsant **plus** an antipsychotic
- Anticonvulsant **plus** an anticonvulsant **plus** an antipsychotic

Third, if response is inadequate, consider ECT for mania with psychosis or catatonia,^d or add clozapine for treatment-refractory illness

Acute Depressive Episode

General Guidelines

- Assess for secondary causes of depression (eg alcohol or drug use)
- Taper off antipsychotics, benzodiazepines, or sedative–hypnotic agents if possible
- Treat substance abuse
- Encourage good nutrition (with regular protein and essential fatty acid intake), exercise, adequate sleep, stress reduction, and psychosocial therapy

Mild to Moderate Depressive Episode

First, initiate and/or optimize mood-stabilizing medication: lithium^a or quetiapine

Alternative anticonvulsants: lamotrigine,^b valproate^a; antipsychotics: fluoxetine/olanzapine combination

Severe Depressive Episode

First, optimize current mood stabilizer or initiate mood-stabilizing medication: lithium^a or quetiapine

Alternative fluoxetine/olanzapine combination

If psychosis is present, initiate an antipsychotic in combination with above

Do not combine antipsychotics

Alternative anticonvulsants: lamotrigine, valproate^a

For severe depressive disorder

Second, if response is inadequate, consider carbamazepine^a or adding antidepressant

Third, if response is inadequate, consider a three-drug combination:

- Lithium **plus** lamotrigine^b **plus** an antidepressant
- Lithium **plus** quetiapine **plus** antidepressant^c

Fourth, if response is inadequate, consider ECT for treatment-refractory illness and depression with psychosis or catatonia^d

Evaluation of Therapeutic Outcomes

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- Pts who have a partial response or non response to therapy should be reassessed for an **accurate diagnosis**, **concomitant medical or psychiatric conditions**, and medications or substances that exacerbate mood symptoms.
- Pts and family members should be actively involved in treatment to monitor target symptoms, response, and side effects

Case study

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- H.M., is a 23-year-old man taken to clinic by his wife. She says that he was doing well until about 3 weeks ago, when his niece was killed in an automobile accident. At the funeral, he borrowed some “nerve pills” from his cousin. Since then, H.M. has been acting increasingly “wild.” He has been staying up later at night and often bursts into the bedroom at 3 am and loudly awakens her.

- Sometimes, he presents her with expensive gifts, which they cannot afford. He often jumps on the bed and starts singing her love songs in a loud voice. He then almost always demands sex, after which he sleeps for a few hours and then loudly gets up and leaves the house. Over the last several weeks, he was noted to be loading his truck in a rapid and reckless manner. His boss received several reports that H.M. was driving unsafe and at high speeds

- Upon arriving at the clinic, H.M. bursts into song, trouble sitting still, listening, or allowing others to talk. His speech is pressured and loud; he often fails to complete sentences or communicate entire ideas. His mood obviously is elevated, but he becomes increasingly irritable throughout the examination. Within moments, he is again smiling and talking of money-making schemes.

1. What factors make H.M. vulnerable to the occurrence of a manic episode at this time?
2. If the “nerve pills” borrowed by H.M. were antidepressants, could they have contributed to the development of his manic episode?
3. What is the appropriate treatment for H.M.'s acute manic episode?
4. Drug of choice for rapid cycling, mixed states, or multiple prior episodes.
5. What are the potential side effects of valproate therapy?

6. How long will it take to get the full effect of lithium?

And what are the monitoring parameter for lithium? Why?

7. After receiving lithium at 1,200 mg/day for 1 year. He is tired and has gained weight in recent weeks and thinks that he is becoming depressed. What is the most likely cause of HM.'s complaints?

8. When BDZ is indicated in bipolar patients?

Case 2 . A.J., a 36-year-old woman, has been maintained successfully on lithium therapy for 5 years for bipolar disorder. She asks whether she should stay on lithium because she plans to become pregnant in the near future.

The End!!!!